"Information Theory"

Topic: Elements of Information Theory :-

Processing
- Source Coding — Try to compress the source
- Channel coding — Try to transmit the info. through channel in efficient manner

Source Coding

Example :- Horse Race

Horse no. :-	1	2	3	4	5	6	7	8
prob. of winning :-	1/2	1/4	1/8	1/16	1/64	1/64	1/64	1/64

We have 365 race so 365 result to be transmitted.

For Fixed length code we need to transmit 3 bit per horse i.e $2^3 = 8$
So to tx the info of winning we need 3 bit on an avg.

For Variable length code we can transmitt least no. of bit for most frequently winning horse. i.e.

Horse no :-	1	2	3	4	5	6	7	8
Prob :-	1/2	1/4	1/8	1/16	1/64	1/64	1/64	1/64
Bit req :-	0	10	110	1110	111100	111101	111110	111111

To decide no. of bits per symbol we need used :- $\log_2 \frac{1}{p_i}$

Avg no. of bits to be transmitted will be :-

$$= \sum_{i=1}^{8} p_i n_i \Rightarrow \frac{1}{2}\times 1 + \frac{1}{4}\times 2 + \frac{1}{8}\times 3 + \frac{1}{16}\times 4 + 4\left[\frac{1}{64}\times 6\right] = 2 \text{ bits}$$

So Avg no. of bits req. in fixed length = 3 bit
Variable " = 2 bit.

Conclusion :-

Prob. of event	Code length	Information in event
Small	long	more
higher	Small	less

1) $I(s_k) = 0 \quad p_k = 1$
2) $I(s_k) \geq 0 \quad 0 \leq p_k \leq 1$
3) $I(s_k) \geq I(s_i) \quad p_k < p_i$
4) $I(s_k, s_i) = I(s_k) + I(s_i)$.

An RV 'X' $\in \{1, 2, 3 \dots M\}$

Amount of Info $= \log_2\frac{1}{p_1}, \log_2\frac{1}{p_2}, \log_2\frac{1}{p_3} \dots \log\frac{1}{p_m}\}$

Code is constructed having code length $\log_2\frac{1}{p_i}$ {no. of bit need will be $\log_2 1/p_i$}

Avj information in the ~~code~~ x is ~~to~~ called Entropy of event.

$$\boxed{H(X) = \sum_i p(i) \log_2 \frac{1}{p(i)}}$$ { Weighted avj. code length. }

<u>Example</u> $X = \begin{cases} 1 & p \\ 0 & 1-p \end{cases}$ (Prob.)

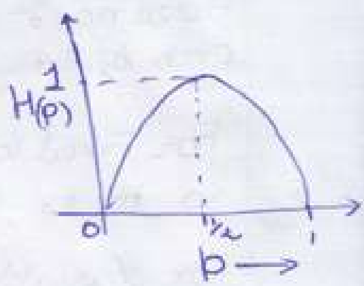
{For proof 8'B' pg.}

$$H(X) = p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$$

$$= -p\log_2 p - (1-p)\log_2(1-p)$$

$$\triangleq H(p) = 0 \quad \text{for } p = 0, 1$$

$$H(p) = 1 \quad \text{for } p = 1/2.$$



$$X = \begin{cases} a & 1/2 & 0 \\ b & 1/4 & 10 \\ c & 1/8 & 110 \\ d & 1/8 & 111 \end{cases}$$

$$H(x) = \frac{1}{2}\log_2 2 + \frac{1}{4}\log_2 4 + \frac{1}{8}\log_2 8 + 1/8 \log 8$$

$$= 7/4 \text{ bits}$$

Max amout of Information will ~~b~~ have 2 bits that is possible when all the prob are equal.

Properties of H(x) equality satisfy when $p(x_i) = 1$

~~or H(x) ≤ log₂ M {M is size of alphabets}~~

$\rightarrow H(x) \geq 0$

Since $P(x_i) \geq 0$

$P(x_i) \leq 1$

$\frac{1}{P(x_i)} \geq 1$

$\log_2 \frac{1}{P(x_i)} \geq 0$

$\sum_i P x_i \log_2 \frac{1}{P(x_i)} \geq 0$

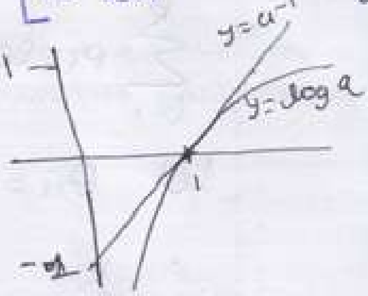
So :- $H(x) \geq 0$

$$\boxed{\begin{array}{c} \text{** Information Rate} \\ R = \text{message rate} \times \text{Entropy} \end{array}}$$

→ $H(x) \le \log_2 K$ [K is size of alphabet X]

Take two PDF $\{p_0, p_1 \cdots p_{k-1}\}$ on alphabet $X = \{S_0, S_1, \cdots S_{k-1}\}$
$\{q_0, q_1 \cdots q_{k-1}\}$

$$\sum_{i=1}^{k} p_i \log_2 \frac{q_i}{p_i} = \frac{1}{\ln 2} \sum p_i \ln \frac{q_i}{p_i}$$ [$\log_2$ is converted to $\ln$]



Using inequality

$$\boxed{\ln a \le a - 1}$$

$$\sum_{i=1}^{k} p_i \log_2 \frac{q_i}{p_i} \le \frac{1}{\ln 2} \sum_{i=1}^{k} p_i \left(\frac{q_i}{p_i} - 1 \right]$$

$$\le \frac{1}{\ln 2} \sum_{i=1}^{k} (q_i - p_i)$$

$$\le \frac{1}{\ln 2} \sum_{i=1}^{k} q_i - \sum_{i=1}^{k} p_i = 0$$

So ∴ $$\sum_{i=1}^{k} p_i \log_2 \frac{q_i}{p_i} \le 0$$

This fundamently inequality holds equality only when $q_i = p_i \forall k$

Taking $q_i = \frac{1}{K}$, $i = 1, 2, 3, \cdots k$

$$\sum_{i=1}^{k} p_i \log_2 \frac{1}{K p_i} = -\sum_{i=1}^{k} p_i \log_2 p_i - \sum_{i=1}^{k} p_i \log_2 K$$

$$= H(X) - \log_2 K \sum_{i=1}^{k} p_i \le 0$$

$$\boxed{= H x \le \log_2 K}$$

Topic Shannons First Theorem :- The source coding theorem:
Given a discrete memory less source of Entropy $H(x)$, the average code word length $\bar{L}$ for any distortionless source encoding scheme is bounded as

$$\bar{L} \ge H(x).$$

$$H(x) \le \bar{L}_{avg} \le H(x) + 1$$

For n symbols

$$n H(x) \le \bar{L} \le n H(x) + 1$$

Proof of $H(x) \leq L_{avg} \leq H(x)+1$

Using Kraft - Mc Millan Inequality.

$$K = \sum_{i=1}^{K} 2^{-l_i} \leq 1 \qquad K \leq 1 \quad \boxed{\log_2 K \leq 0}$$

$$\sum_{i=1}^{K} p_i \log_2 \frac{Q_i}{p_i} \leq 0$$ equality holds for $p_i = Q_i$

Let $Q_i = \frac{2^{-l_i}}{K} \qquad K = \sum_{i=1}^{K} 2^{-l_i}$

$$\sum_{i=1}^{K} Q_i = \frac{1}{K} \sum_{i=1}^{K} 2^{-l_i} = 1$$

$$\sum_{i=1}^{K} p_i \log_2 \frac{2^{-l_i}}{K p_i} = \sum_{i=1}^{K} p_i \left[\log_2 \frac{1}{p_i} - l_i - \log_2 K\right]$$

$(p_i l_i = L_{av})$

$$= H(x) - L_{avj} - \log_2 K \leq 0 \qquad \{\log_2 K \leq 0\}$$

$$= H(x) - L_{avj} \leq \log_2 K \leq 0$$

$$\boxed{= L \geq H(x)} \qquad H(x) - L_{avj} \leq 0$$

Using $\log_2 \frac{1}{p_i} \leq l_i \leq -\log_2 \frac{1}{p_i} + 1$

multipling with p_i and summing over i yelds.

$$= -\sum_{i=1}^{K} p_i \log_2 p_i \leq \sum_{i=1}^{K} p_i l_i \leq \sum_{i=1}^{K} p_i \left[\log_2 \frac{1}{p_i} + 1\right]$$

$$\boxed{= H(x) \leq L_{avj} \leq H(x) + 1}$$

Topic Entropy Coding :- The design of Variable length Code such that its average codeword length approaches the entropy of DMS is ~~or~~ ~~reffered~~ refered to as entropy Coding

→ Shannon Fano Coding

→ Huffman Coding

Properties of prefix Codes :- 1) $H(x) \leq \bar{L} \leq H(x)+1$

2) $\sum_{k=0}^{K-1} 2^{-l_k} \leq 1$ To be satisfied.

→ Shannon Fano Coding : Shannon Fano Algorithm :

→ List the source symbols in order of decreasing probability

→ Partition the set into two sets that are (sums of prob) close to equiprobables as possible and assign 0 to the uper and 1 to the lower set.

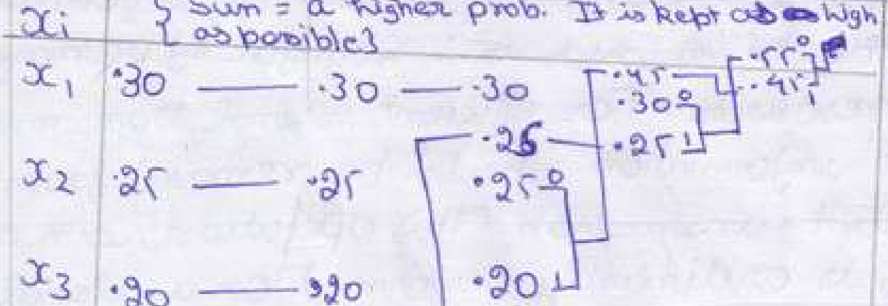
→ Continue this process, each time partitioning the set with as nearly equal probabilities as possible until further partitioning is not possible.

x_i	$P(x_i)$	Step 1	Step 2	Step 3	Step 4	Code.
x_1	·30	0	0			00
x_2	·25	0	1			01
x_3	·20	1	0			10
x_4	·12	1	1	0		110
x_5	·08	1	1	1	0	1110
x_6	·05	1	1	1	1	1111

$$H(x) = \sum_{i=1}^{6} p_i \log_2 \frac{1}{p_i}$$
$$= 2.36 \text{ bits/Symbol}$$
$$L = \sum_{i=1}^{k} p_i l_i$$
$$= 2.38 \text{ bits/Symbol}$$
$$\eta = \frac{H(x)}{L} \times 100 = 99\%$$

→ Huffman coding

{Sum = a higher prob. It is kept as high as possible}



·30 – 00
·25 – 10
·20 – 11
·12 – 011
·08 – 0100
·05 – 0101

$$H(x) = \sum_{i=1}^{k} p_i \log_2 \frac{1}{p_i} = 2.36 \text{ bit}$$

$L = \sum p_i n_i$ ~~d~~ $= 2.38$ bits

$\eta = 0.99$.

AHAP	ALAP
→ $H(x) = 2.36$	$H(x) = 2.36$
$L = 2.38$	$L = 2.38$
$\eta = .99$	$\eta = .99$

$$\sigma^2 = \sum_{i=1}^{k} p_i [n_i - \bar{L}]^2$$

↑ Bit per Symbol.

AHAP

$\sigma^2 = .16$ $\qquad \sigma^2 = 1.36$

So:.

AHAP is the Best

Lempel Ziv Coding :

Data → 000101110010100101

Numerical position	Code book Dictionary location	Shortest subsequence	Representation	Code word.
1	0001	0	01	0000
2	0010	1	02	0001
3	0011	00	$\boxed{1}$1	0010
4	0100	01	$\boxed{1}$2	0011
5	0101	011	$\boxed{4}$2	1001
6	0110	10	$\boxed{2}$1	0100
7	0111	010	$\boxed{4}$1	1000
8	1000	100	$\boxed{6}$1	0110
9	1001	101	$\boxed{6}$2	0111

4 → Binary equivt. 100; 2 → data 2 → 1 → $\boxed{1001}$

→ Does not require the probabilities for generating Code word

→ It is good for very long haul of data

→ Efficiency is good when data rates are high [mbps, gbps ...]

Data Compaction :- The data generated by the source contain ~~sufficient~~ significant amount of information that is redundant, the tx of which is ∴ wasteful of primary communication resources. For efficient signal transmission the redundant information should be removed from the signal prior to transmission. This operation, with no loss of information, is ordinarily performed on a signal in digital form, in which case we refer to it a data compaction or lossless data compression.

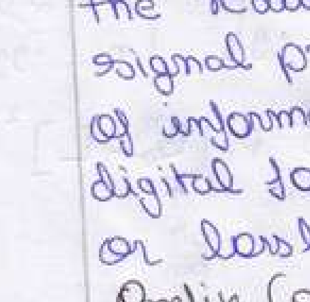
Prefix Coding

Source Symbol	Prob	Code 1	Code 2	Code 3
a	1/2	0	0	0
b	1/4	1	10	01
c	1/8	00	110	011
d	1/8	11	111	0111

Code 1 → no reason. $\sum_{l=1}^{k} 2^{-l_k} \le$ is not satisfied.

Code 3 → No code for 'a' is p... to all other co...

Code 2 → yes satisfies both c...



$H(X) \le \bar{L} \le H(X)+1$ & $\sum_{k=1}^{K-1} 2^{-l_k} \le 1$

Joint Entropy & Conditional Entropy :-

Joint Entropy $\Rightarrow H(X,Y) = \sum_{x}\sum_{y} p(x,y) \log \frac{1}{p(x,y)}$

$H(X,Y) \neq H(X) + H(Y)$ { in general }

Example $X = Y$

$H(X) = H(Y) = H(X,Y)$ { Y & X have same information }

But for Independent X & Y

$H(X,Y) = H(X) + H(Y)$

To prove :-

$X \in (0 \quad 1)$ $Y \in (0 \quad 1)$

prob :- $p \quad 1-p$ prob $q \quad 1-q$.

$$H(X,Y) = \sum_{X}\sum_{Y} p(x,y) \log_2 \frac{1}{p(x,y)}$$

x \ y	0	1
0	pq	$p(1-q)$
1	$(1-p)q$	$(1-p)(1-q)$

$$= pq \log \frac{1}{pq} + (1-q)p \log \frac{1}{p(1-q)} + (1-p)q \log \frac{1}{(1-p)q}$$

$$+ (1-p)(1-q) \log_2 \frac{1}{(1-p)(1-q)}$$

$$= p\left[q\left(\log\frac{1}{p} + \log\frac{1}{q}\right) + (1-q)\left(\log\frac{1}{p} + \log\frac{1}{(1-q)}\right)\right]$$

$$+ (1-p)\left[q\left(\log\frac{1}{q} + \log\frac{1}{(1-p)}\right) + (1-q)\left(\log\frac{1}{(1-p)} + \log\frac{1}{1-q}\right)\right]$$

$$= p\left[q \log\frac{1}{p} + q \log\frac{1}{q} + (1-q)\log\frac{1}{p} + (1-q)\log\frac{1}{1-q}\right]$$

$$+ (1-p)\left[q \log\frac{1}{q} + q \log\frac{1}{1-p} + (1-q)\log\frac{1}{(1-p)} + (1-q)\log\frac{1}{1-q}\right]$$

$$= \boxed{p \log\frac{1}{p}} + q \log\frac{1}{q} + \boxed{(1-p)\log\frac{1}{(1-p)}} + (1-q)\log\frac{1}{(1-q)}$$

$$= \boxed{H(X)} + H(Y)$$

$H(X,Y) = H(X) + H(Y)$

Conditional Entropies

$$H[Y|X=x] = -\sum_{y} p[Y=y|X=x] \log [p\,Y=y|X=$$

$$H[Y|X] = -\sum_{x} p(x)\, H[Y|X=x]$$ ⇐ Conditional Entropy.

$$H[Y|X] = -\sum_{x}\sum_{y} p(x,y] \log p\left[\frac{y_i}{x_i}\right]$$

$$= -E_{p(x,y)} \log (p\, y/x]$$

$\boxed{p(x_i y_i) = p(x_i) p}$

Relation :-

1> $H(x,y) = H(x) + H(y/x)$
$\qquad = H(y) + H(x/y)$

2> $H(x) \geq H(x/y]$ & $H(y) \geq H(y/x)$

3> $H(x,y) \leq H(x) + H(y)$.

Proof 1 :- $H(x,y) = \sum_{x}\sum_{y} p(x,y) \log \frac{1}{p(x,y)}$

$$= -\sum_{x}\sum_{y} p(x,y) \log p(x) - \sum_{x}\sum_{y} p(x,y) \log p\left(\frac{y}{x}\right]$$

$$= H(x) - \sum_{x}\sum_{y} p(x,y) \log p\left(\frac{y}{x}\right]$$

$$= H(x) + H(y/x]$$

Proof 2 :- $H(x) - H(x/y) = \sum_{x} p(x) \log \frac{1}{p(x)} + \sum_{x}\sum_{y} p(x,y) \log p($

$$= \sum_{x}\sum_{y} p(x,y) \log_2 \left(\frac{p(x/y)}{p(x)}\right)$$

Using $\ln(a) \geq 1 - a$

$$LHS \geq \sum_{x}\sum_{y} p(x,y) \left[1 - \right]$$

$$\geq \sum_{x}\sum_{y} p(x,y)\left[1 - \frac{p(x,y)}{p(x)p(y)}\right]$$

Proof 2 :- $H(X) \geq H(X/y)$

$$H(x) - H\left(\frac{x}{y}\right) = \sum_x p(x) \log \frac{1}{p(x)} + \sum_x \sum_y p(x,y) \log p\left(\frac{x}{y}\right)$$

$$\ln \frac{1}{a} \geq 1 - a.$$

$$LHS \geq \sum_x \sum_y p(x,y) \left[1 - \frac{p(x)}{p(x/y)}\right]$$

$$\geq \sum_x \sum_y p(x,y) \left[1 - \frac{p(x)p(y)}{p(x,y)}\right] \qquad p(x,y) = p_y\left(p\frac{x}{y}\right)$$

$$\geq \sum_x \sum_y p(x,y) - p(x)p(y)$$

$$\geq \sum_x p(x) - p(x)$$

$$\geq 0$$

$$\boxed{H(x) \geq H\left(\frac{x}{y}\right)}$$

Qus Find $H(x)$, $H(y)$, $H(x/y)$, $H(y/x)$, $H(x,y)$

$P(x,y) =$

y \ x	1	2	3	4
1	1/8	1/16	1/32	1/32
2	1/16	1/8	1/32	1/32
3	1/16	1/16	1/16	1/16
4	1/4	0	0	0

$P(x_1) = 1/2$ $P(x_2) = \frac{1}{4}$ $P(x_3) = \frac{1}{8}$ $P(x_4) = 1/8$ $H(x) = \sum_{i=1}^{4} P x_i \log_2 \frac{1}{P x_i} = \frac{7}{4}$

$P(y_1) = \frac{1}{4}$ $P(y_2) = 1/4$ $P(y_3) = 1/4$ $P(y_4) = 1/4$ $H(y) = \sum_{i=1}^{4} P(y_i) \log_2 \frac{1}{P(y_i)} = 2$

$$H(x,y) = \sum\sum P(x,y) \log \frac{1}{P(x,y)}$$

$$= 2\left[\frac{1}{8} \log_2 8\right] + 6\left[\frac{1}{16} \log_2 16\right] + 4\left[\frac{1}{32} \log_2 32\right] + \frac{1}{4} \log_2 4$$

$$= \frac{27}{8} \text{ bits.}$$

$$H\left(\frac{X}{Y}\right) = \sum\sum P(x,y)\log_2 P\left(\frac{x_i}{y_i}\right)$$

$$= \sum_{y_i} P(y_i)\, H(X/y_i)$$

$$= \frac{1}{4}\left[H\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right]\right.$$

$$+ \frac{1}{4}\left[H\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{8}, \frac{1}{8}\right]\right.$$

$$+ \frac{1}{4}\left[H\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]\right.$$

$$+ \frac{1}{4}\left[H\left[1, 0, 0, 0\right]\right.$$

$$= \left[\frac{1}{4}\left[\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}\right]\right] \times 2 + \frac{1}{4} \times 2 + \frac{1}{4} \times 0$$

$$= \frac{7}{8} + \frac{1}{2} = \frac{11}{8} \text{ bits.}$$

$$H\left(\frac{Y}{X}\right) = \sum_{y_i} P(x_i)\, H(Y/x_i)$$

$$= \frac{1}{2}\left[H\left[\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{2}\right]\right.$$

$$+ \frac{1}{4}\left[H\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0\right]\right.$$

$$+ \frac{1}{8}\left[H\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0\right]\right.$$

$$+ \frac{1}{8}\left[H\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0\right]\right.$$

$$= \frac{1}{2}\left[\frac{1}{2} + \frac{3}{8} + \frac{3}{8} + \frac{1}{2}\right] + \frac{1}{4}\left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right]$$

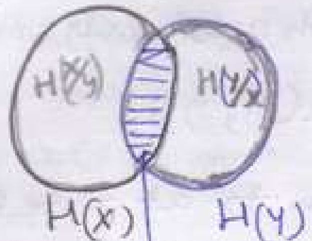
$$+ \frac{1}{8}\left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right] + \frac{1}{8}\left[\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right]$$

$$= \frac{13}{8}$$

$$H[X,Y] = H(Y) + H\left(\frac{X}{Y}\right) = 2 + \frac{11}{8} = \frac{27}{8}$$

$$= H(X) + H\left(\frac{Y}{X}\right) = 7/4 + \frac{13}{8} = \frac{27}{8}$$

Representation as Ven diagram.



Common / mutal Info b/ω X & y

$$I[X;y] = H(x) - H\left(\frac{x}{y}\right) = H(y) - H(y/x)$$

Mutual Information:

$$I[X;y] = \sum_x \sum_y p(x,y) \log_2 \frac{p(x,y)}{p(x)\,p(y)}$$

If X & y are independent $I(X;y] = 0$

$$I[X;y] = H(x) - H(X|y)$$

$$= \sum_x \sum_y p(x,y) \log_2 p(x|y) - \sum_x \sum_y p(x,y) \log_2 p(x)$$

$$= -H(X/y) + H(x)$$

$$= H(x) - H(X/y).$$

Properties of Mutual Information.

(i) $I(X,y] = I(y,X)$

(ii) $I[X;Y) \geq 0$

(iii) $I[X;Y) = H(Y) - H(Y/X]$

(iv) $I[X;y] = H(X) + H(Y) - H(X,Y]$

Topic **Channel Coding Theorem: Shannon's Second:-**

In terms of channel capacity C is stated in two parts:-

1> Let a DMS with an alphabet X having entropy H(X) and produces symbols every T_s seconds, let DMC having capacity C and be used once every T_c second, Then if

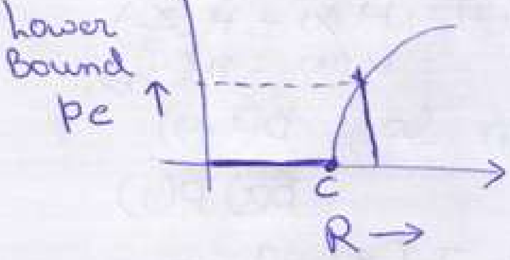
$$\frac{H(X)}{T_s} \leq \frac{C}{T_c} \rightarrow \text{critical rate}$$

There exists a coding scheme for which the source ~~coding~~ o/p can be tx over the channel and be reconstructed with an arbitrarily small **pe**. If equality is satisfied then system is said to be signaling at the critical rate.

ii) Conversely if $\frac{H(x)}{T_s} > \frac{C}{T_c}$ it is not possible to tx information over the channel and reconstruct it with arbitrarily small p_e.

Conclusion: $C \triangleq \max_{P(x)} I[X;Y]$

For any $\epsilon > 0$ & $R < C$, we can tx at rate R with $P_e < \epsilon$
(R ↑ Rate)



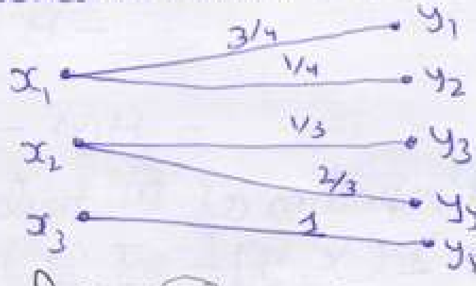
If $R > C$ then "reliable Communication" is not possible.

Topic: Types of Channels :→

Lossless Channel :

A channel described by a channel matrix with only one nonzero element in each column is called a lossless channel.

$$P(y|x) = \begin{bmatrix} 3/4 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

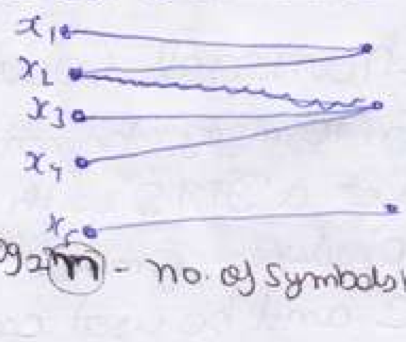


$C \triangleq \max_{P(x)} H(X) = \log_2 m$ - no. of symbol in x

Deterministic Channel.

A channel described by a channel matrix with only one non-zero element in each row is called Deterministic channel.

$$P(y|x) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

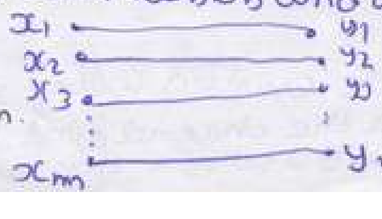


$C_s \triangleq \max_{P(x)} H(Y) = \log_2 m$ - no. of Symbols in y

Noise less channel.

A channel is called noiseless if it is both lossless and deterministic

$C \triangleq \max_{P(x)} H(X) = \max_{P(x)} H(Y) = \log_2 n$
$= \log_2 m$.



Binary Symmetric Channel.

A BSC channel has two I/P ($x_1=0$, $x_2=1$) & two O/P ($y_1=0, y_2=1$)
This channel is symetric because the probability of receiving a 1 is a 0 is sent is the same as prob. of receiving 0 if one is sent.

$$C = 1 + p\log_2 p + (1-p)\log_2 p$$

Topic: Information Capacity Theorem : Shannon's Third Theorem:-

Stated as:- The information capacity of a continous channel of bandwidth B hz, perturbed by additive white Gaussian noise of PSD $\frac{N_0}{2}$ and limited in BW to B is given by

$$C = B\log_2\left[1 + \frac{P}{N_0 B}\right) \text{ bits per second.}$$

where P is avj transmitted power.

Ques Considering a binary memory less source X with two symbols x_1 & x_2. Find the max value of entropy. Or Show that max value of entropy occurs when they are equi probable by first principle.

$$P_1(x) = p \quad \text{and} \quad P_2(x) = (1-p)$$

$$H(x) = -p \log p - (1-p) \log (1-p) \quad -1$$

$$\frac{d}{dp} H(x) = -\left[\frac{d}{dp}[p \log p] + \frac{d}{dp}(1-p)\log(1-p)\right]$$

$$\left[\frac{d}{dx} u.v = u\frac{dv}{dx} + v\frac{du}{dx}\right]$$

$$= -\left[p \times \frac{1}{p} + \log p \times 1 + (1-p) \times \frac{-1}{(1-p)} + \log(1-p)(-1)\right] \left[\log(1-p) = \frac{-1}{(1-p)}\right]$$

$$= -[1 + \log p - 1 - \log(1-p)]$$

$$= \log(1-p) - \log p = \log\left(\frac{1-p}{p}\right)$$

$$\frac{d}{dp} H(x) = 0 \quad \text{for max value.}$$

$$\log\left(\frac{1-p}{p}\right) = 0$$

Taking anti log:

$$\frac{1-p}{p} = 1$$

$$1 - p = p$$

$$\boxed{p = \frac{1}{2}}$$

For $p = \frac{1}{2}$ we get max entropy. putting the value in 1

$$H(x) = -\frac{1}{2}\log\frac{1}{2} - \left(\frac{1}{2}\right)\log\frac{1}{2} = 1 \text{ bit/Symbol.}$$

"ERROR CONTROL CODING"

Topic: Important Terms Related to Coding :-

Code Word :- The code word is the n bit encoded block of bits. It contains message bits and parity or redundant bits.

Code Rate :- The code rate is defined as the ratio of the number of message bits (k) to the total number of bits (n) in a code word $(r) = \frac{k}{n}$

Hamming Weight and Distance :-

Hamming Weight of a code word C of length n is defined as W(C) i.e the no. of non zero element in the code.

Hamming distance of two code words a & b is defined as d(a,b), is the no. of ~~of~~ position in which a & b differ.

Relation b/w hamming weight & distance is

$$W_C = d[C, 0]$$

$$d(a,b) = w(a \oplus b)$$

Minimum distance, Error detection and Correction Capabilities :-

Theorem 1 : The minimum distance dmin of a linear code C is the smalles Hamming weight of the non zero codeword in the C.

Theorem 2 : A linear code C of minimum distance dmin can detect upto 't' errors if and only if

$$\boxed{d_{min} \geq t+1}$$

Theorem 3 : A linear code C of minimum distance dmin can ~~detect~~ correct upto t errors if and only if

$$d_{min} \geq 2t+1$$

⇒ Linear Block Codes :→

1) We consider an (n, k) LBC $n \to$ Codeword length
$k \to$ message bits
$c = n-k \leftarrow$ redundant bits.

2) The code vector is represented by

$$X = [M : C]_{1 \times n}$$

$M = k$ messagebits vector.
$C = (n-k)$ parity vector

3) A block code generator generates the parity vectors required to be added to the message bits to generate the code words.

$$X = MG$$

X = Code Vector of $1 \times n$ size
M = message " " $1 \times k$ "
G = Generator matrix of $k \times n$ size.

4) The Generator matrix is dependent on the type of LBC used. It is Generally represented as

$$[G] = [I_k | P]$$

$I_k = k \times k$ identity matrix
$P = k \times (n-k)$ Coefficient matrix.

5) The parity vector can be obtained as

$$[C]_{1 \times (n-k)} = [M]_{1 \times k} [P]_{k \times (n-k)}$$

Example:
(6,3) LBC

$$G = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

$n = 6$
$k = 3$
$c = 3$

Find all Code vector.

$$I(k) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$P_{(k)} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}_{3 \times 3}.$$

SNO	Message Vector	Parity bits	Complete code Vecor.
1	0 0 0	0 0 0	000 000
2	0 0 1	1 1 0	001 110
3	0 1 0	1 0 1	010 101
4	0 1 1	0 1 1	011 011
5	1 0 0	0 1 1	100 011
6	1 0 1	1 0 1	101 101
7	1 1 0	1 1 0	110 110
8	1 1 1	0 0 0	111 000

$$[C_0\, C_1\, C_2] = [m_0\, m_1\, m_2] \begin{bmatrix} P_{00} & P_{10} & P_{20} \\ P_{01} & P_{11} & P_{21} \\ P_{02} & P_{12} & P_{22} \end{bmatrix}$$

$$C_0 = m_0 P_{00} \oplus m_1 P_{01} \oplus m_2 P_{02}$$

$$C_1 = m_0 P_{10} \oplus m_1 P_{11} \oplus m_2 P_{12}$$

$$C_2 = m_0 P_{20} \oplus m_1 P_{21} \oplus m_2 P_{22}$$

Topic Hamming Codes:→

These are again LBC, $[n,k)$. The family (n,k) Hamming code for $q \geq 3$ is defined by.

→ $n = 2^q - 1$ $\quad q$ = no. of parity bits i.e $(n-k)$

→ $k = 2^q - q - 1$ ← no of message bits

→ no. of parity bits $(n-k) = q$

$q \geq 3$ i.e minimum no. of parity bits is 3

H matrix is also used in LBCs i.e

$$[H] = [P^T \,\vdots\, I_{n-k}]$$

$$= [P^T]_{(n-k)\times k}$$

H is called parity check matrix.

Example The parity check matrix of a (7, 4) LBC is given

$$[H] = \begin{bmatrix} 1110 & \vdots & 100 \\ 1101 & \vdots & 010 \\ 1011 & \vdots & 001 \end{bmatrix}$$

$$G = [I_k \vdots P]$$

$$H = [P^T \vdots I_{n-k}]$$

$$[P^T] = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \quad [P] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{4\times 3}$$

$$G = \begin{bmatrix} 1000 & \vdots & 111 \\ 0100 & \vdots & 110 \\ 0010 & \vdots & 101 \\ 0001 & \vdots & 011 \end{bmatrix}$$

$d_{min} = 3$

$3 \geq S+1$

$\boxed{S \leq 2}$ no of error detect

$t \leq 1$ " " " Correct

<u>Topic</u> Syndrome Decoding for Block code :-

$$[S] = Y[H^T]$$

$$[S]_{1\times(n-k)} = [Y]_{1\times n} [H^T]_{n\times(n-k)}$$

Example
$$[H] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$y = 0110110$ Find S

$$[S] = 0110110 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

0110110
0010000
0100110

$[S] = [101] = 0010000$

$y \oplus S_e =$

With Syndrome decoding, an (n,k) LBC can correct upto t errors per codeword if n & k satisfy the following Hamming Bound

$$2^{n-k} \geq \sum_{i=0}^{t} \binom{n}{i}$$

$$\binom{n}{i} = \frac{n!}{(n-1)!\ i!}$$

A block code for which equality holds is known as the perfect code. Single error correcting perfect codes are called Hamming Codes.

Ques For $(6,3)$ Systematic LBC, the 3 parity-check bits C_4, C_5 & C_6 are formed from following

$$C_4 = d_1 \oplus d_3$$
$$C_5 = d_1 \oplus d_2 \oplus d_3$$
$$C_6 = d_1 \oplus d_2$$

a) Find G

b) Construct all possible code word

c) $Y = [010111]$ Correct the error.

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$G = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

S No	Message $m_0\ m_1\ m_2$	Parity	Code
1	0 0 0	0 0 0	000 000
2	0 0 1	1 1 0	001 110
3	0 1 0	0 1 1	010 011
4	0 1 1	1 0 1	011 101
5	1 0 0	1 1 1	100 111
6	1 0 1	0 0 1	101 001
7	1 1 0	1 0 0	110 100
8	1 1 1	0 1 0	111 010

$$H = [P^T \vdots I_k]$$

$$= \begin{bmatrix} 1 & 0 & 1 & \vdots & 1 & 0 & 0 \\ 1 & 1 & 1 & \vdots & 0 & 1 & 0 \\ 1 & 1 & 0 & \vdots & 0 & 0 & 1 \end{bmatrix} \qquad H^T = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[S] = 010111 \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$y = 0\,1\,0\,1\,0\,1\,1$$

$$C = 0\,0\,0\,1\,0\,0$$

$$\overline{\quad 0\,1\,0\,0\,1\,1 \quad}$$

$$[S] = [1\ 0\ 0] \quad 4 \text{ bit}$$

Q = For $(n,1)$ Block Code. There are only 2 code word in reception code , an all 0 code word and an all 1 code word. Consider the case of a reception code with $n=5$.

→ Construct the Generator matrix G for this $[5,1]$ block code

→ Using G find all code words

→ Find the parity check matrix H for this Code.

→ Show that $GH^T = 0$

a) $n = 5 \quad k = 1 \qquad P = [1\ 1\ 1\ 1]$

$$G = [1 \vdots 1\ 1\ 1\ 1]$$

b) $C_1 = [0][1\ 1\ 1\ 1\ 1] = [0\ 0\ 0\ 0\ 0]$

$C_2 = [1][1\ 1\ 1\ 1\ 1] = [1\ 1\ 1\ 1\ 1]$

c) $H = [P^T \vdots I_k]$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

d: $$GH^T = [1\ 1\ 1\ 1\ 1] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= [0\ 0\ 0\ 0]$$

Q. Consider a single-error correcting code for 11 data bits?

a) How many check bits are required?

b) Find a parity-check matrix H for this code.

$$2^{n-k} \geqslant \sum_{i=0}^{1} \binom{n}{i} = \binom{n}{0} + \binom{n}{1} = 1 + n$$

$$\left\{ \begin{array}{c} n-k=m \\ k=11 \\ n=m+11 \end{array} \right\} \quad 2^m \geqslant 12+m \rightarrow 4 \leq m$$

at least 4 check bits are required.

$$H = [P^T : I_k]$$

$$\begin{array}{c} \begin{array}{ccccccccccccccc} & & & & & & & & & & & 8 & 4 & 2 & 1 \end{array} \\ = \left[\begin{array}{ccccccccccc:cccc} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \end{array}$$

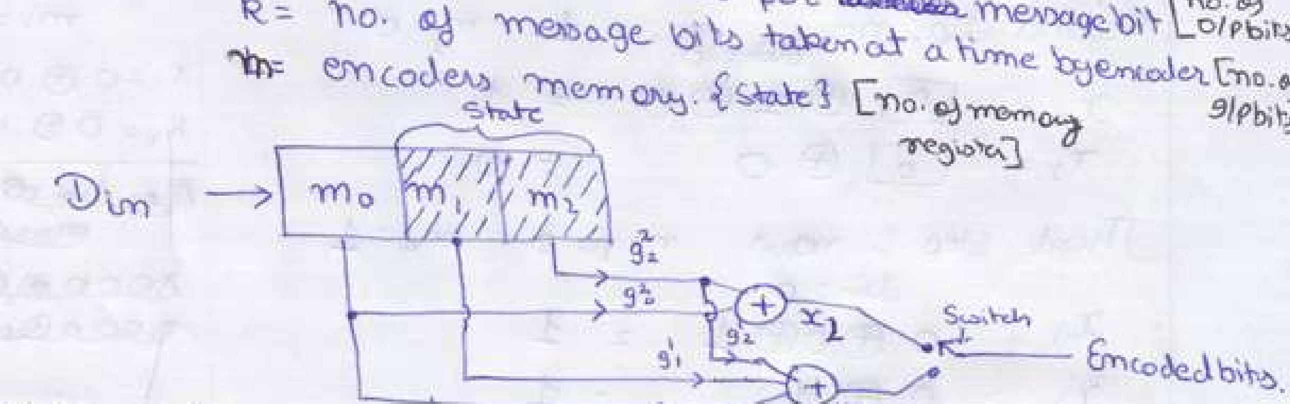
Topic: Convolution Codes.

Block Codes	Convolution Codes (also called Recurrent codes)
→ n Code word length K message bits. Redundancy is introduced Block by Block	→ Redundancy is introduced Bit by Bit.
→ There are fixed and finite no. of ways to code 2^k Codewords	→ We doesn't consider any linear combination of codeword. Total no. of code is define acc. to constraint length.
→ dmin is calculated	dfree is calculated.
→ (n, k) Linear Codes	{n, k, L} Convolution Code.

Encoder for Convolution Coding:-

(2, 1, 2) Convolution encoder. $[n[\overset{\text{no. of message bits}}{K}+L-1]]$

(n, k, L)
- n = no. of encoded bits per message bit [no. of O/P bits]
- k = no. of message bits taken at a time by encoder [no. of I/P bits]
- m = encoders memory. {State} [no. of memory registers]



{n ⇒ x_1 x_2
k ⇒ m_0
m ⇒ m_1, m_2}

(2, 1, 2)

approx: [2, 1, 3] ◄ 2 (m k L)

$\boxed{L = k[m+1]}$.

Code Rate ⇒ $\frac{k}{n} = r$

Constraint length $L = n(m+1)$

$x_2 = g_0^2 m_0 \oplus g_1^2 m_2$

$x_1 = g_0^1 m_0 \oplus g_1^1 m_1 \oplus g_2^1 m_2$

Example:- Encode (10001) given generators arg for Top order (1 1 1) and for lower order (1 0 1). Construct a Convolution Code

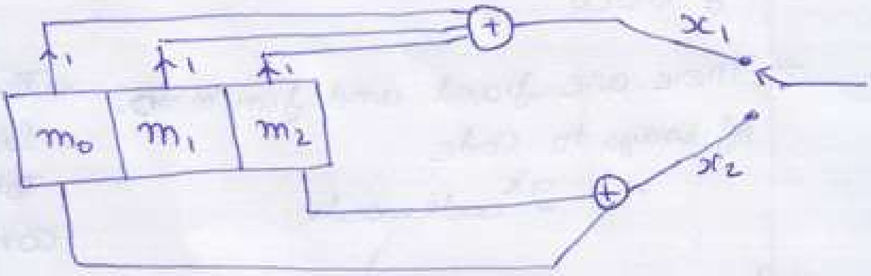
$$[g_0^1\ g_1^1\ g_2^1] = [1\ 1\ 1] \qquad [g_0^2\ g_1^2\ g_2^2] = 101$$

So ∴ encoder will be

$L = n[m+1]$

$= 2[3]$

$= 6$



$d_5 d_4 d_3 d_2 d_1 = [1\,0\,0\,0\,1]$

Ist step: Initially $m_1 = m_2 = 0$

$x_1 = \boxed{1} \oplus 0 \oplus 0 = 1$ (din)

$x_2 = \boxed{1} \oplus 0 = 1$

Second step: now din = 0, $m_1 = 1$, $m_2 = 0$

$x_1 = \boxed{0} \oplus 1 \oplus 0 = 1$

$x_2 = \boxed{0} \oplus 0 = 0$

Third step: now din = 0, $m_1 = 0$, $m_2 = 1$

$x_1 = 0 \oplus 0 \oplus 1 = 1$

$x_2 = 0 \oplus 1 = 1$

IV Step now din = 0, $m_1 = 0$, $m_2 = 0$

$x(1) = 0 \oplus 0 \oplus 0 = 0$

$x(2) = 0 \oplus 0 = 0$

V step din = 1, $m_1 = 0$, $m_2 = 0$

$x(1) = 1 \oplus 0 \oplus 0 = 1$

$x(2) = 1 \oplus 0 = 1$

VI Step din 1, $m_1 = 1$, $m_2 = 0$

$x(1) = 1 \oplus 1 \oplus 0 = 0$

$x(2) = 1 \oplus 0 = 1$

VII din = 0, $m_1 = 1$, m_2 1

$x_1 = 0 \oplus 1 \oplus 1 = 0$

$x_2 = 0 \oplus 1 = 1$

VIII din 0, $m_1 = 0$, m_2 1

$x_1 = 0 \oplus 0 \oplus 1 = 1$

$x_2 = 0 \oplus 1 = 1$

Codeword =

11 01 01 11 00 11 10

0 0 1 1 0 0 0 1

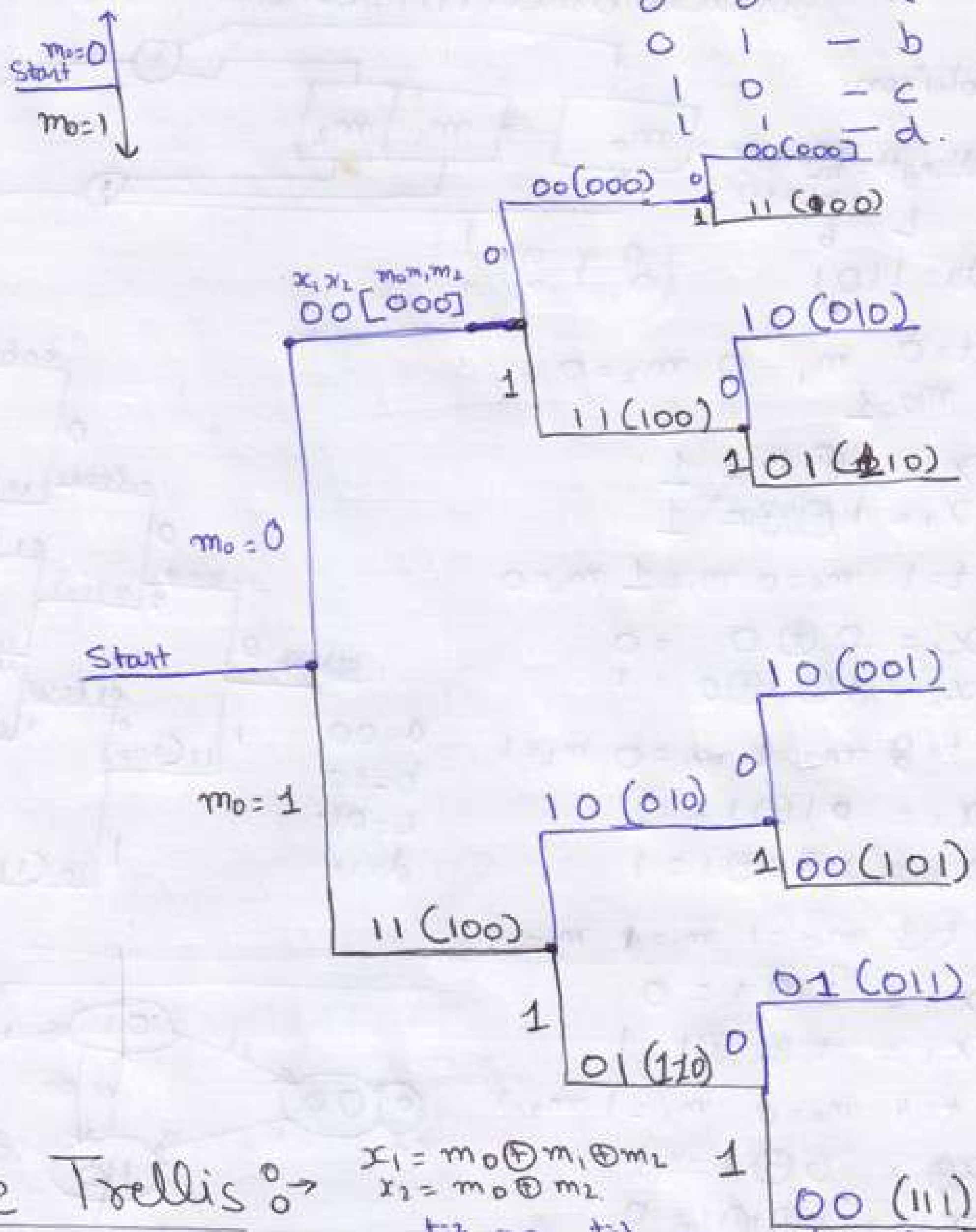
$2[6+2]$

16

The Code Tree

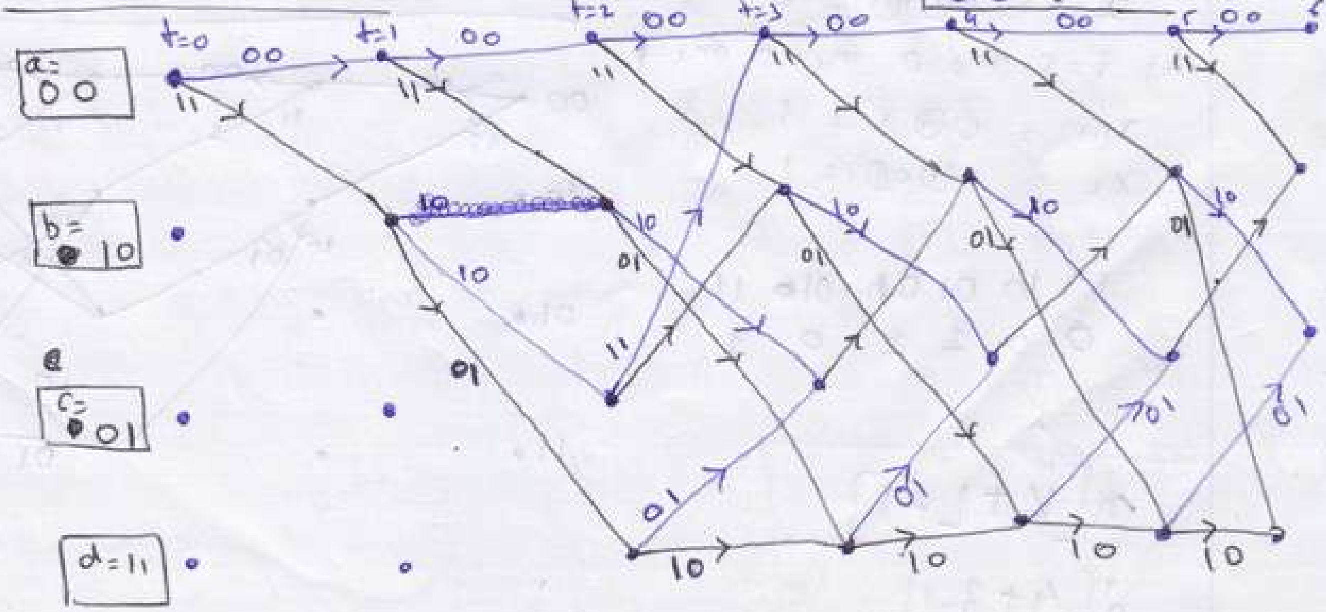
States $m_1\ m_2$

0	0	– a
0	1	– b
1	0	– c
1	1	– d.



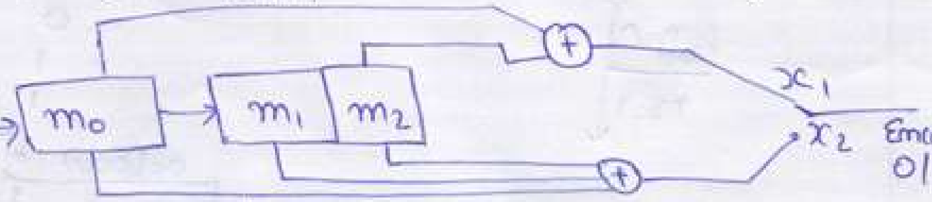
Code Trellis :

$x_1 = m_0 \oplus m_1 \oplus m_2$
$x_2 = m_0 \oplus m_2$



Example Encode 1101 given generator are Top order (101) and for lower order (111). Construct. Trellis diag. and calc Tree.

Solution.



word length = n[m+1]
= n[2+1]
L = 6

data = 1101 (2, 1, 2)

At t=0 $m_1 = 0$ $m_2 = 0$
$m_0 = 1$

$x_1 = 1 \oplus 0 = 1$
$x_2 = 1 \oplus 0 \oplus 0 = 1$

At t=1 $m_0 = 0$ $m_1 = 1$ $m_2 = 0$

$x_1 = 0 \oplus 0 = 0$
$x_2 = 0 \oplus 1 \oplus 0 = 1$

At t=2 $m_0 = 1$ $m_1 = 0$ $m_2 = 1$

$x_1 = 1 \oplus 1 = 0$
$x_2 = 1 \oplus 0 \oplus 1 = 1$

At t=3 $m_0 = 1$ $m_1 = 1$ $m_2 = 1$

$x_1 = 1 \oplus 1 = 0$
$x_2 = 1 \oplus 1 \oplus 1 = 1$

At t=4 $m_0 = 0$ $m_1 = 1$ $m_2 = 1$

$x_1 = 0 \oplus 1 = 1$
$x_2 = 0 \oplus 1 \oplus 1 = 0$

At t=5 $m_0 = 0$ $m_1 = 0$ $m_2 = 1$

$x_1 = 0 \oplus 1 = 1$
$x_2 = 0 \oplus 0 \oplus 1 = 1$

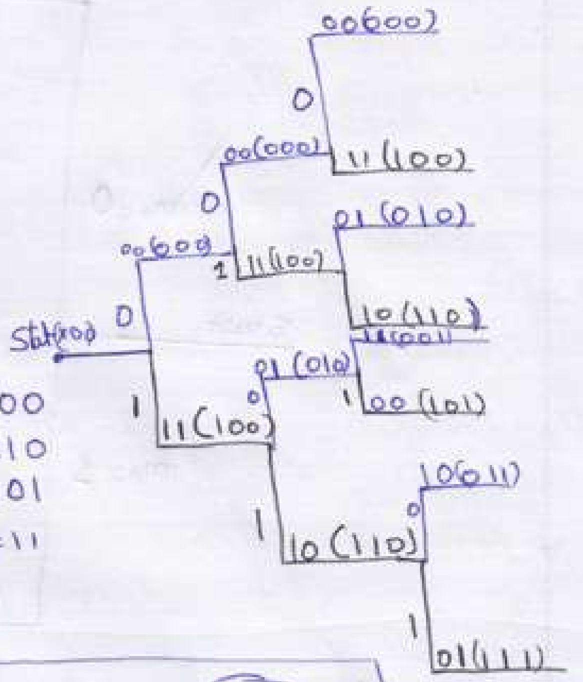
11 10 01 00 01 11
0 0 1 1 0 1

K

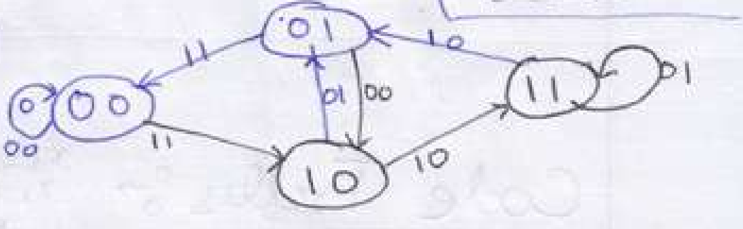
$n[K + L - 1]$

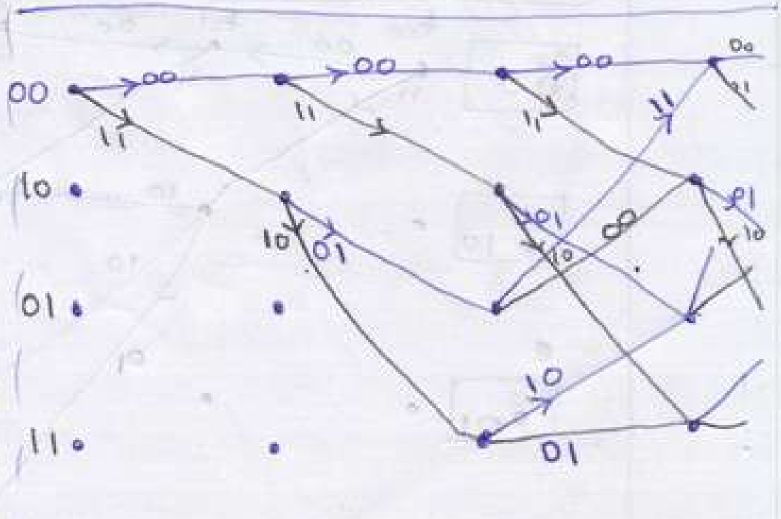
$2[4 + 3 - 1]$

$2[6] = 12$



a = 00
b = 10
c = 01
d = 11





Decoding of Convolutional Codes :-

Viterbi Algorithm :->

→ Algorithm operats on the principle of maximum likelihood decoding and achieves optimum performance.

→ The maximum likelihood decoder has to examin the entire received sequence Y and find a valid path which has smallest Hamming distance from M.

→ out of 2^N paths, maximum likely hood principle with be sort the paths with minimum hammig distance,

→ Those path will make maximum likelihood decoding posible.

Metric → It is defined as the Hamming distance of each branch of each surviving path from the corresponding branch of Y [received Signal]

Surviving path :- The surviving path is defined as the path of the decoded signal with minimum metric.

Free Distance & Coding Gain :-

D_{Free} → Is the minimum weight of path that deviates from all zero path and later merges back into the all zero path at some point further down the trellis diagram.

Code Gains ⇒ $A = \frac{r d_{free}}{2} = \left(\frac{E_b}{N_o}\right)$

r = Code rate

$$d_{free} \geq 2t + 1$$

t = error to be corrected.

Advantages :-

1> highly Satisfactory bit error performance

2> High speed of operation

3> Ease of implementation

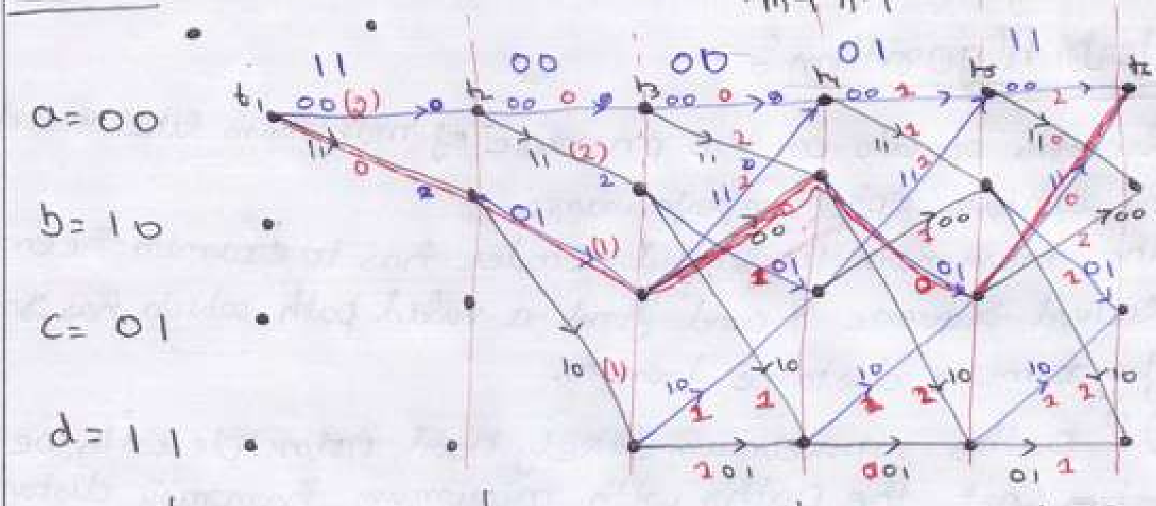
4> Low cost.

no. of msg bits

$$t = (K + n - 1)$$

Example :- $y = 11\ 00\ 00\ 01\ 11\ 00\ 00\ 0$

$m + n - 1$



SNo	Paths	Path matric	Decision	Dfree.
1	$a_1\ b_2\ c_3\ a_4$	$0 + 1 + 2 = 3$	×	
2	$a_1\ b_2\ d_3\ c_4\ a_5$	$0 + 1 + 1 + 1 + 1 = 4$	×	
3	$a_1\ b_2\ d_3\ d_4\ c_5\ a_6$	$0 + 1 + 1 + 2 + 0 = 4$	×	
4	$a_1\ b_2\ c_3\ b_4\ c_5\ a_6$	$0 + 1 + 0 + 0 + 0 = 1$	✓	

Decoded $\boxed{10100}$

For dfree.

Sno.	Paths	Path Bits.	O/P.	W.
1	$a_1 b_2\ c_3\ a_4$	100	11 0 111	5 ✓
		1100	11 10 10 11	6
2	$a_1 b_2\ d_3\ c_4\ a_5$		11 10 01 10 11	7
3	$a_1\ b_2\ d_3 d_4 c_5\ a_6$	11100		
4	$a_1 b_2 c_3\ b_4 c_5 a_6$	10100	11 01 00 01 11	6

dfree = 5

14

$= n[5 + 2]$ (14) $\boxed{M + n - 1}$

$= 2[$

Topic

Cyclic Code :-

A code is cyclic if

1> C is a linear Code

2> any cyclic shift of a code word is also a code word i.e if the codeword $a_0 a_1 \ldots a_{n-1}$ is a 'C' then $a_{n-1} a_0 \ldots a_{n-2}$ is also 'C'

Generator Polynomial for the Cyclic Code :-

$$X(p) = M(p) \cdot G(p)$$

$X(p)$ ↓ Code word polynomial; $M(p)$ ↓ Message polynomial; $G(p)$ ↓ Generator polynomial.

$$G(p) = 1 + g_1 p + g_2 p^2 + \ldots + g_{n-k-1} p^{n-k-1} + p^{n-k}$$

Theorem If $G(p)$ is a polynomial of degree n-k and it is the factor of $p^n + 1$. Then $G(p)$ generates an (n, k) cyclic code word in which code polynomial $X(p)$ for data Vector $M(p)$ is given by

$$X(p) = M(p)\, G(p)$$

* Generation of Non Systematic Code Words

$$X_1(p) = M_1(p) \; G(p)$$
$$X_2(p) = M_2(p) \; G(p)$$

These all generate a non systematic code word.

* For Systematic Code Word.

i) We multiply $M(p)$ by p^{n-k} to get $p^{n-k} M(p)$. This multiplication is equivalent to shifting the message bits by (n-k) bits.

ii) We divide the shifted message polynomial $p^{n-k}M(p)$ by $G(p)$ to obtain $C(p)$

$$R(p) = \frac{p^{n-k}\, M(p)}{G(p)}$$

iii) $R(p)$ is added to shifted message polynomial $G(p)$ to obtain the Code word polynomial $X(p) = [p^{(n-k)} M(p)] \oplus R(p)$.

Example Consider a (7,4) cyclic code with $g(x) = 1+x+x^3$

a) Let $d = (1010)$. Find Corresponding Codeword.

b) Let $c = [1100101]$ Find " data word.

a)

$G(x) = 1 + x + x^3$

$d = 1 \quad 0 \quad 1 \quad 0$

$D(x) = 1 + 0x + 1x^2 + 0x^3$

$= 1 + x^2$

$C(x) = (1+x^2)\,[1+x+x^3]$

$= 1 + x + x^3 + x^2 + x^3 + x^5$

$= 1 + x + x^2 + x^5$

$$\boxed{C = 1\ 1\ 1\ 0\ 0\ 1\ 0}$$

For Systematic Codeword.

$G(x) = 1 + x + x^3$

$D(x) = 1 + x^2$

$x^{n-k} = x^3$

$R(x) \Rightarrow \dfrac{x^3(1+x^2)}{(1+x+x^3)}$

$$x^3+x+1 \,\big)\, x^5 + x^3 \,\big(\, x^2$$

$$x^5 + x^3 + x^2$$

$$x^2$$

$R(x) = 001$

$C(x) = x^3(1+x^2) \oplus x^2$

$= 0001010$

0010000

$\boxed{001} \boxed{1010}$

Parity Message.

b) $C = 1100101$

$C(x) = 1 + x + x^4 + x^6$

$G(x) = 1 + x + x^3$

$D(x) = \dfrac{C(x)}{G(x)}$

$$x^3+x+1 \,\big)\, x^6 + x^4 + x + 1 \,\big(\, x^3 + 1$$

$$x^6 + x^4 + x^3$$

$$x^3 + x + 1$$

$$x^3 + x + 1$$

$$x$$

$D(x) = 1 + x^3$

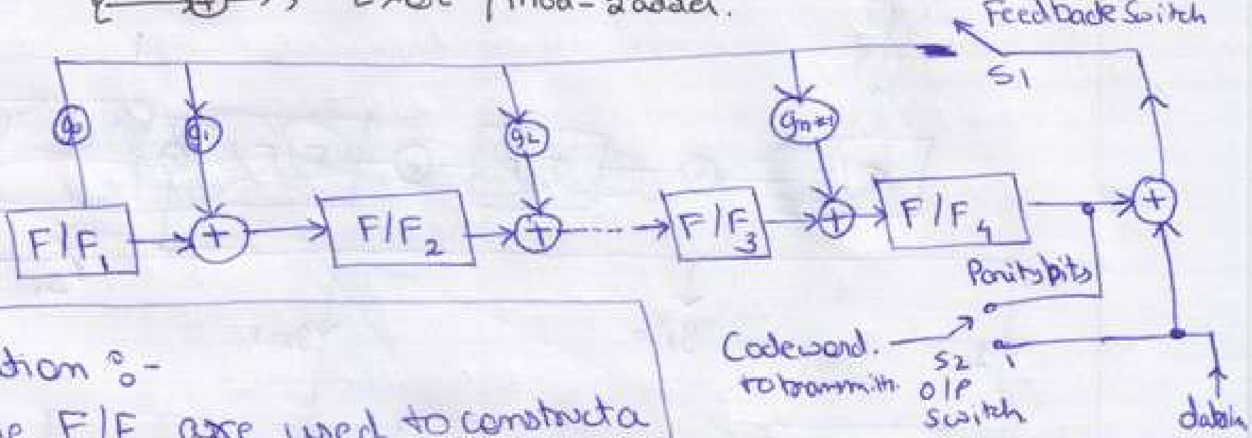
$d = [1001]$

→ Encoder of Cyclic Codes :-

Symbols → [F/F] → Flip-Flops = 

{ → (g_1) → ; → (g_2) → } Multiplying Coefficients.

{ → (+) → } Exor / mod-2 adder.



Operation :-

→ The F/F are used to construct a shift register. The content of F/F will get shifted in the direction given by arrow corresponding to each clock pulse.

→ The Feedback Switch (S_1) is closed and O/P switch is connected to ~~message~~ data In. All F/F are at zero initially. First k message are shifted to the tx and also shifted into shift register.

→ After shifting the k message bits the shift register will contain the (n-k) parity or check bits.

→ After shifting k message bits S_1 = Open S_2 is set to parity bits O/P. Now every shift will give parity bit O/P.

"Equations."

$$Q_1 = d g_0 \oplus Q_4$$

$$Q_2 = d g_1 \oplus Q_1$$

$$Q_3 = Q_2 \oplus d g_2 \ldots$$

$\vdots$

✱ right most bit of message will go first.

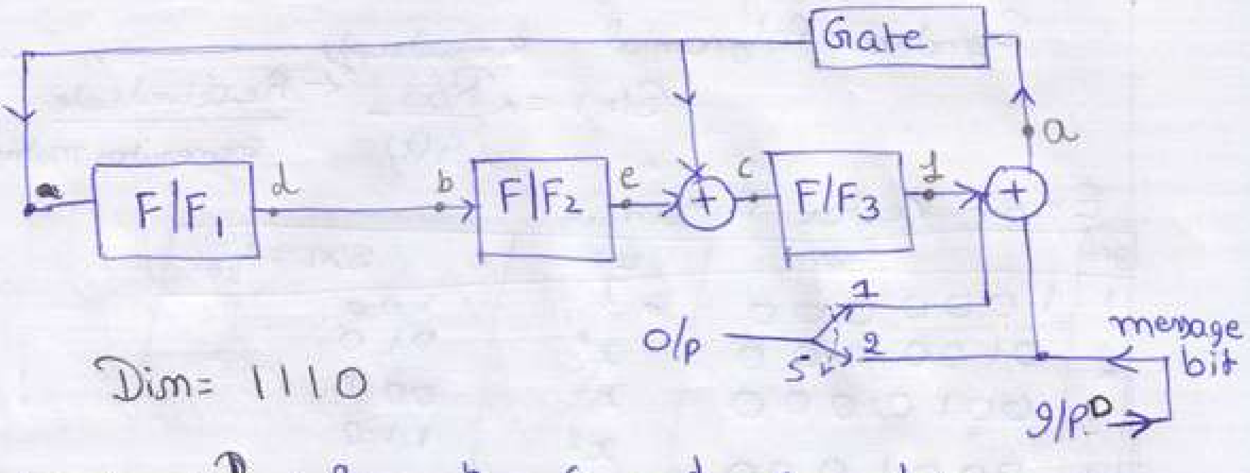
Ques Example Construct a (7,4) cyclic code ~~g(x)~~

$g(x) = 1 + x^2 + x^3$

$d = 1110$

Draw a Encoder circuit too?

$g(0) = 1 \quad g(1) = 0 \quad g(2) = 1 \quad g(3) = 1$



Dim = 1110

Shift no	Gate	S=	Din (D)	g/Ps F/F1 (a)	F/F2 (b)	F/F3 (c)	O/Ps F/F1 (d)	F/F2 (e)	F/F3 (f)	Codeword at O/P
0	ON	2	0	0	0	0	0	0	0	0
1	ON	2	1	$D \oplus f$ 1	b = d 0	$a \oplus e$ 1	0	0	0	10
2	ON	2	1	0	1	0	1	0	1	110
3	ON	2	1	1	0	0	0	1	0	$\boxed{1110}$ ← d
4	OFF	1	~~0~~	0	1	0	1	0	0	F/F3 $\boxed{0}$ $\boxed{1110}$
5	"	1	~~0~~	0	0	1	0	1	0	$\boxed{00}$ $\boxed{1110}$
6	"	1	~~0~~	0	0	0	0	0	1	$\boxed{100}$ $\boxed{1110}$

Dim = 1001 St = 3 till

$$\left\{\begin{aligned} a &= D \oplus f \\ b &= d \\ c &= a \oplus e \end{aligned}\right\} t = t_i$$

Shift no	Gate	S=	Din (D)	g/Ps F/F1 (a)	F/F2 (b)	F/F3 (c)	O/Ps F/F1 (d)	F/F2 (e)	F/F3 (f)	Codeword at O/P
0	ON	2	1	1	0	1	0	0	0	1
1	ON	2	0	1	1	1	1	0	1	01.
2	ON	2	0	1	1	0	1	1	1	001
3	ON	2	1	1	1	0	1	1	0	$\boxed{1001}$
4	OFF	1	0	0	1	1	1	1	0	0 $\boxed{1001}$
5	off	1	0	0	0	1	0	1	1	10 $\boxed{1001}$
6	off	1	0	0	0	0	0	0	1	110 $\boxed{1001}$

3

Syndrome Decoding for the Cyclic Code

Syndrom Polynomial

$$S(x) = \frac{R(x)}{G(x)}$$

Remainder of; ← Received code; Generator matrix

Error Vector Pattern is

S.No	E(x)	E x	$S(x) = \frac{E(x)}{g(x)}$
1	1000000	1	100
2	0100000	x	010
3	0010000	x^2	001
4	0001000	x^3	110
5	0000100	x^4	011
6	0000010	x^5	111
7	0000001	x^6	101

$S_1(x) = \frac{E_1(x)}{g(x)} : \frac{1}{1+x+x^3}$

$x^3+x+1 \overline{) \; 1 \;} \; (1+1$
x^3+x+1
x^3+x
x^3+x+1
$\boxed{1}$

$S_2(x) = \frac{E_2(x)}{G(x)} = \frac{x}{1+x+x^3}$

$1+x+x^3 \overline{) \; x \;} \; (1+1$
$1+x+x^3$
x^3+1
x^3+1+x
$\boxed{x}$

$S_3 = x^3+x+1 \overline{) \; x^2 \;} \; (1+x$
$x^3+0+x+1$
x^3+x^2+x+1
$x^3+0+x+1$
$\boxed{x^2}$

$S_4 = x^3+x+1 \overline{) \; x^3 \;} \; (1$
x^3+x^0+1
$\boxed{x+1}$

$S_5 = x^3+x+1 \overline{) \; x^4 \;} \; (x$
x^4+x^2+x
$\boxed{x^2+x}$

$S_6 = x^3+x+1 \overline{) \; x^5 \;} \; (x^2+1$
$x^5+x^3+x^2$
x^3+x^2
x^3+x+1
$\boxed{x+x^2+1}$

$S_7 = x^3+x+1 \overline{) \; x^6 \;} \; (x^3+x+1$
x^6
x^4+x^3
x^4+x^2+x
x^3+x^2+x $\boxed{x^2+1}$
x^3+x+1
x^2+1

Example Decode 0110010 find Error. $G(x) = 1+x+x^3$

$R(x) = x + x^2 + x^5$

$G(x) = 1 + x + x^3$

$$S(x) = x^3+x+1 \overline{) x^5 + x^2 + x} \; (x^2+1$$
$$\underline{x^5 + x^3 + x^2}$$
$$x^3 + x$$
$$\underline{x^3 + x + 1}$$
$$\underline{1}$$

$S(x) = 100$

$E(x) = 1000000$

$Y(x) = R(x) \oplus E(x) = 0110010$
1000000
$\boxed{1110010}$

Topic Boss - Chaudhuri - Hocquenghem [BCH] Codes:-

→ BCH code is one of the most powerfull known class of Linear cyclic block codes.

→ BCH are known for their multiple error correcting ability and the ease of encoding and decoding

→★ The difference b/w BCH and normal cyclic code is that in cyclic we construct a Code and then find out its minimum distance in order to estimate its error correcting capabilities. But in BCH, we will start from other end i.e we begin by specifying the no. of random error we desire the code to correct. Then we go on to construct the generator polynomial for the code.

The most common BCH codes are charaterized by:-

Block length: $n = 2^m - 1$

No. of message bits: $k = n - mt$

Minimum distance: $d_{min} \geqslant 2t+1$

{ For any positive integer m [with $m \geqslant 3$]) and $t < \frac{(2^m - 1)}{2}$ }

This type of BCH codes are called Primitive BCH codes.
→ The Block lengths and code rate of BCH Codes are Variable.

Generator Polynomials of BCH Codes.

n	k	t	Generator Polynomial.
7	4	1	1 011
15	11	1	10 011
15	7	2	111 010 001
31	26	1	100 101
31	21	2	11 101 101 001
31	16	3	1 000 111 110 101 111
31	11	5	101 100 010 011 011 010 101

Topic: Reed Solomon Codes :-

→ RS codes are an Important subset of the non-Binary BCH with a wide range of applications in digital communicat- and data Storage. The typical applications area of the RS codes are

- Storage devices {including CD, B-R Disk, DVD}
- Wireless or mobile communication.
- Satellite Communication.
- Digital television
- High Speed modems such as those employing ADSL, X DSL etc.

→ RS codes are particularly good at dealing with bursts of error: Six consecutive bit errors, for example can affect at most two bytes. Thus even a double error correction version of a RS code can provide a comfortable safty factor.

→ Currently RS code in CD tech. are able to cope with error burst as long as 4000 consecutive bits.

For RS codes most popular value of m is 8. The RS code with m=8 is extremely powerfull.

The parameter of a t-error correcting RS code may be listed as under

Block length : $n = 2^m - 1$

Message size : k = symbols

Parity check Bit size : $n - k = 2t$ Symbols

Minimum distance : $d_{min} = (2t+1)$ Symbols.

Advantages of R-S Code :-

1) The RS codes make the highly efficient use of redundancy.
2) It is possible to adjust the block lengths and the symbol size to accomodate a wide range of message sizes. [Highly BW efficient]
3) The RS codes provide a wide range of code rate. The code rate can be chosen to obtain the optimum performance.
4) The efficient decoding tech. are avialable for decoding the RS codes.

Some parameters for RS codes acc to

$$n = 2^m - 1$$

k: Symbols

Parity check bit :- $n - k = 2t$

$d_{min} = (2t + 1)$

Code rate $= \frac{k}{n}$

$n = 2^m - 1 = 15$
$n - k = 2$
$15 - k = 2$
$k = 13$

m	$q = 2^m$	$n = q - 1$	t	k	d^*_{min}	$r = k/n$
2	4	n = 3	1	1	3	0.333
3	8	7	1	5	3	0.7143
			2	3	5	0.4286
			3	1	7	0.1429
4	16	15	1	13	3	.866
			2	11	5	.733
			3	9	7	.600
			4	7	9	.4667
			5	5	11	.333
			6	3	13	.200
			7	1	15	.0667
5	32	31	1	29	3	.935
			5	21	11	.6774
			8	15	17	.4839
8	256	255	5	245	11	.9608
			15	225	31	.8824
			50	155	101	.6078

Topic

Trellis Code Modulation :-

Introduction :-

→ In error control coding technique we add extra bits to the information bits in an known manner. This is used to improve bit error rate.

→ This improved "BER" is at the cost of Bandwidth. This BW expansion is equal to the reciprocal of the code rate.

For eg. RS[255,233]

$$R = \frac{233}{255} = \cdot 875 \quad \frac{1}{R} = 1\cdot 143.$$

→ Hence to send 1000 bits 143 extra bits (overhead) is to be transmitted. This means that we need extra 14.3% BW to transmit 1000 bits. So we can observe that even a highly efficient code need expansion of such large BW.

→ In power limited channels we may trade the BW expansion for a desired performance. But for BW limited channels (like TCN) this may not be an easy option. For such channel PAM, QAM, QPSK, MPSK are used to support High BW.

So ∴ we can conclude that for good BER we have to ~~tradeoff~~ sacrifice BW or Power. [To prove this we can use $C = B\log_2(1+SNR)$, C is fixed, B & SNR can be varied.]

→ To optimize the above discussed problem we introduces the optimum modulation scheme that is known as

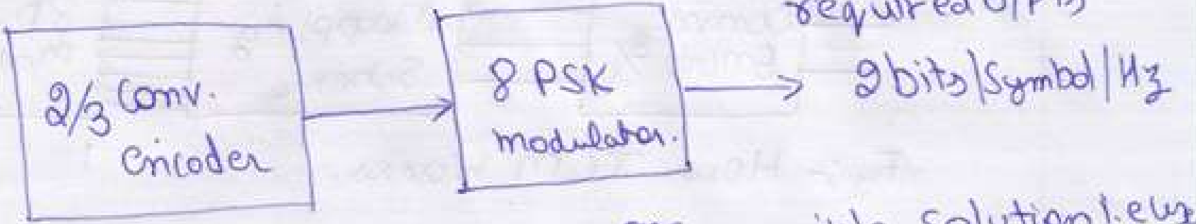
$\boxed{\text{CODED MODULATION SCHEME}}$

Concept of Coded Modulation :-

→ We optimize coder and Modulator Independently even though they are used for same purpose i.e to lower the BER. and than we sacrifice BW or Power for that.

→ ~~In~~ To avoide sacrificing of BW or Power we can optimize Both coder and Modulator simultaneously i.e ~~by doing~~ ~~incorporate~~ if channel encoder is integrated with modulator.

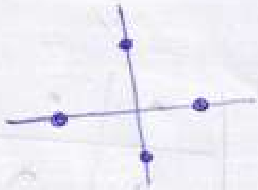
Consider a Communication System givenbelow :-



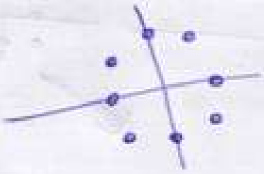
→ To achive required O/P we have one possible Solution i.e. use Uncoded QPSK.

→ Another possiblity is to first use a Conv Code with $r = \frac{2}{3}$ [which conver 2 uncoded bits to 3 coded bits] and then use an 8psk signal set which has through put of 3bits/s/Hz. This 8 PSK will give the same O/P as of QPSK i.e 2 bits/s/Hz.

→ But there is a problem in doing so. !



QPSK



8 PSK

Distance B/w the two symbols is more ~~[illegible]~~ than better (Euclidean Distance) is the system performance.

more distance — less interference

less distance — more interference.

→ So we can say that Usin 8 PSK will Compensate the advantage of using $\frac{2}{3}$ conv. encoder.

→ So to use this Scheme we need to increase the Code gain of convolution encoder such that it outweights the performance loss because of 8-PSK signal set. If this is achieve then we can say that improvement in BER is achieved without sacrificing BW or Power.

→ In this example we have used Trellis encoder with the modulator so it called TCM. Schem.

TCM Trellis Code Modulator.

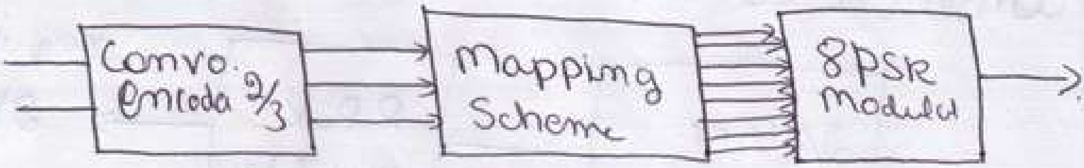
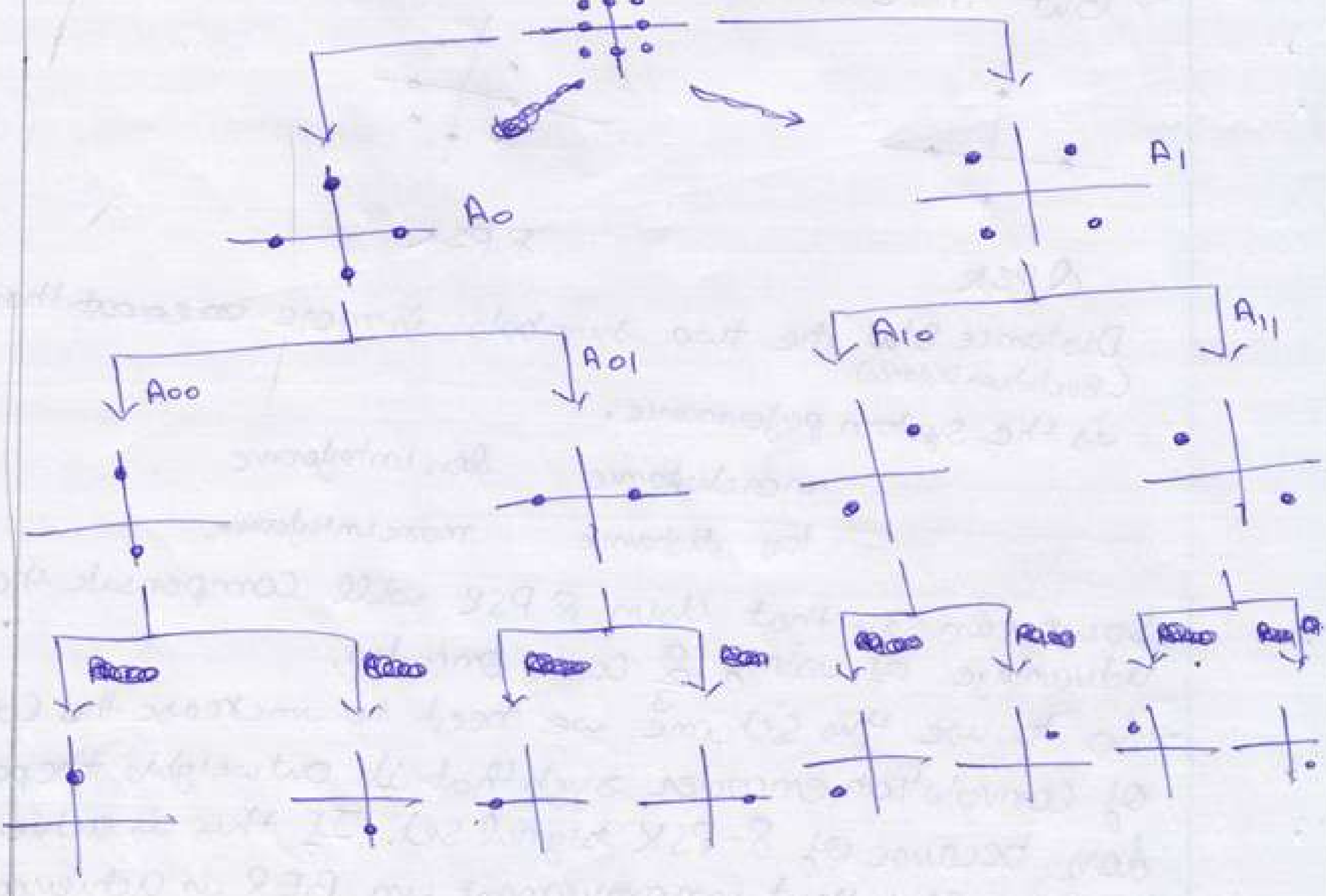


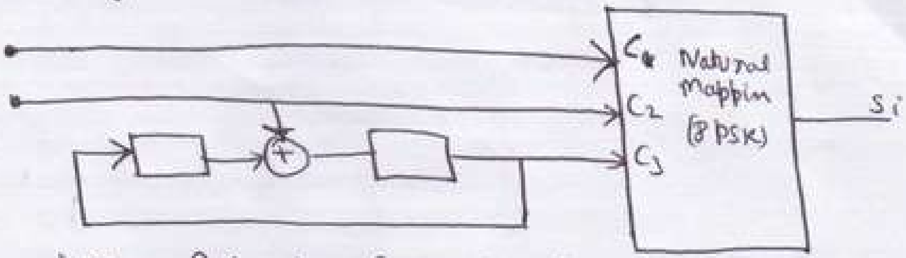
Fig :- How TCM Works.

Mapping :- "mapping by Set Partitioning.
The mapping by set partitioning is based on successive partitioning of the expanded 2^{n+1}-ary signal set into subsets with increasing minimum Euclidean distance. Each time we partition the set, we reduce the numer of the signal points in the subset, but increasing the minimum distance b/w the signal in the subset.

For eg :- Consider the set partitioning of 8PSK



→ Block Diagram for TCM



using Recursive Convo. encoder.

→ Example → TCM scheme that involves 16QAM. TCM encoder takes in 3 bits and o/p one symbol from 16QAM constellation diagram.